

Nonreal Roots Survive in Polynomial Iterative Equations

A cubic counterexample and a classification of circular spectra

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Abstract

For a real polynomial $P(z) = \sum_{j=0}^d a_j z^j$, write $P[f](x) = \sum_{j=0}^d a_j f^{\circ j}(x)$, where $f^{\circ 0}(x) = x$. A published question of Draga and Morawiec asks whether nonreal characteristic roots of maximal modulus can be removed when a real root has the same modulus. We give a negative answer under the stated continuity assumptions. An explicit, odd, increasing, globally bi-Lipschitz homeomorphism F satisfies $F^{\circ 3}(x) = 8x$, although $F(1) = 3$. Its minimal iterative polynomial is $z^3 - 8$, certified by a 3×3 Hankel determinant equal to -56 . A second construction is real analytic at every real point and has minimal polynomial $(z - 1)^2(z^2 + 1)$; deleting its nonreal roots also fails.

We then determine exactly which monic real polynomials supported on one circle $|z| = r > 0$ occur as minimal iterative polynomials of increasing homeomorphisms of $\mathbb{R}$. For $r \neq 1$, the positive root r must have odd multiplicity, at least as large as every other multiplicity. For $r = 1$, the identity is exceptional; otherwise the multiplicity of 1 must be even and strictly larger than every other multiplicity. Every permitted polynomial is realized by an explicit small trigonometric perturbation of a polynomial-exponential coordinate. The proof uses two-sided orbit recurrences and an elementary positivity lemma. We also obtain sharp increasing-solution rigidity criteria and explain the role of differentiability at a fixed point. Exact rational verification code, certificates, and the random area-selection record accompany the article.

Status. The counterexamples and classification are proved below. This is an AI-assisted, unrefereed research draft, not a proof-assistant-checked publication. A targeted literature search located the published open question but did not establish priority for the answer or for the classification. The conjugacy mechanism itself is not claimed to be new.

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1 The problem selected and the scope of the answer

1.1 A precise published target

Let $I \subseteq \mathbb{R}$ be a nondegenerate interval and let $f : I \rightarrow I$ be continuous. An equation

$$a_d f^{\circ d}(x) + a_{d-1} f^{\circ d-1}(x) + \cdots + a_1 f(x) + a_0 x = 0 \quad (1)$$

with $a_0 a_d \neq 0$ is a polynomial-like iterative equation. Its *characteristic polynomial* is $P(z) = \sum_{j=0}^d a_j z^j$. The notation concerns composition, not pointwise powers: in general, $f^{\circ 2}(x)$ is not $f(x)^2$.

The specific target is Problem 6.1 of Draga and Morawiec [1], in the last section of their paper. In the context of lowering the order of (1), they ask about deleting nonreal roots of largest modulus when a real root has that same modulus. The question also appears in the authors' preprint, page 13. Here *deleting roots* means replacing P by the corresponding polynomial divisor while preserving the implication that every solution satisfies the lower-order equation.

The answer under these hypotheses alone is negative. In fact, the cubic counterexample in Section 2 satisfies no nonzero equation of lower order whatsoever. The analytic counterexample in Section 3 shows that lack of smoothness is not the general reason for failure.

The equal-modulus qualification is essential. Existing order-reduction results employ strict separation of root moduli, together with orbit monotonicity or alternating monotonicity; Draga's later note discusses additional elimination cases and again identifies the unrestricted nonreal-root issue as unresolved [2]. Our examples do not contradict those theorems: they lie at the spectral boundary excluded by strict separation.

1.2 Relation to the initial candidate

The literature search began with Petrov's MathOverflow question asking for conditions on P that force a continuous solution of $P[f] = 0$ to be linear [3]. The retrieved question had no posted answer. It led to the more sharply formulated root-elimination problem above.

Throughout this article, *homogeneous linear* means $f(x) = cx$ and *affine* means $f(x) = cx + b$. The distinction matters when 1 is a multiple root. We do not claim a complete answer to Petrov's question for arbitrary polynomials or for both monotonicity orientations. We do obtain an exact answer for *increasing solutions when all characteristic roots lie on one circle*; see Section 8.

1.3 Random selection, exclusion, and priority

Before choosing a problem, a list of 80 mathematical areas was supplied to an operating-system-backed random generator through `secrets.randbelow(80)`. The first draw returned the zero-based index 17: area 18, *Functional equations*. There was no redraw. The complete ordered list and the result are preserved in `data/selection.json`.

The supplied `manifest(1).tex` was reviewed in full. The present problem was not selected from its entries. Its SHA-256 fingerprint is recorded with the selection so that the exclusion check can be tied to the actual supplied version; see Appendix B.

The target is therefore an explicitly published open question, not a new question invented for this report. The search found the original statement, subsequent related papers, and no explicit later answer. That is not an exhaustive bibliographic certification that the question remained open up to the preparation date. The

mathematical conclusion below is independent of that priority issue: the proposed general elimination is false as printed. Nor should the elementary conjugacy construction be advertised as a new technique. The contribution developed here is the explicit counterexample, its connection with the published question, and the self-contained spectral classification.

2 An exact cubic counterexample

2.1 The two-branch definition

Define a function on one fundamental interval by

$$\phi(u) = \begin{cases} (u+5)/2, & 1 \leq u \leq 3, \\ 4u-8, & 3 \leq u \leq 8. \end{cases} \quad (2)$$

The formulas agree at $u = 3$. For $x > 0$, choose the unique $k \in \mathbb{Z}$ such that $8^k \leq x < 8^{k+1}$, and put

$$F(x) = 8^k \phi(8^{-k}x). \quad (3)$$

Complete the definition by

$$F(0) = 0, \quad F(-x) = -F(x) \quad (x > 0). \quad (4)$$

The endpoint identity $\phi(8) = 24 = 8\phi(1)$ makes the scaling construction consistent. It gives, on the whole real line,

$$F(8x) = 8F(x). \quad (5)$$

Theorem 2.1 (Cubic counterexample). *The function F defined by (2)–(4) is an odd, strictly increasing homeomorphism of $\mathbb{R}$ with*

$$\frac{1}{2}|x-y| \leq |F(x) - F(y)| \leq 4|x-y| \quad (x, y \in \mathbb{R}). \quad (6)$$

It satisfies

$$F^{\circ 3}(x) = 8x \quad (x \in \mathbb{R}), \quad (7)$$

but is not homogeneous linear or affine. The smallest degree of a nonzero polynomial Q with $Q[F] = 0$ is three, and the monic polynomial of that degree is

$$\mu_F(z) = z^3 - 8 = (z-2)(z^2 + 2z + 4). \quad (8)$$

Proof. On every scaled copy of the two intervals in (2), the slope of F is either $1/2$ or 4 . The pieces meet continuously. Moreover,

$$\frac{4}{3} \leq \frac{F(x)}{x} \leq 3 \quad (x > 0). \quad (9)$$

For the first branch this follows from $\phi(u)/u = 1/2 + 5/(2u)$; for the second it follows from $\phi(u)/u = 4 - 8/u$. Thus $F(x) \rightarrow 0$ as $x \rightarrow 0$, and $F(x) \rightarrow \infty$ as $x \rightarrow \infty$. Oddness handles the negative half-line. Strict monotonicity and surjectivity follow.

Between two strictly positive points there are only finitely many breakpoints, so their secant slope lies between $1/2$ and 4 . Pass to a limit when one endpoint is zero, and use oddness and addition across zero to obtain (6) for all pairs of points.

To check the iterate relation it suffices to work on $[1, 8]$. There are three relevant itinerary intervals, because an intermediate iterate crosses a breakpoint at $x = 4$. Direct substitution gives Table 1. In using the table, values in $[8, 64]$ are computed with (5). The final column is $8x$ in every row. Scaling extends the identity to every positive x ; oddness and the value at zero finish the proof of (7).

The values $F(1) = 3$ and $F(3) = 4$ show that F is not homogeneous linear. Since $F(0) = 0$, it is not affine either. Minimality is proved immediately below. $\square$

Table 1: Exact branchwise verification of the cubic relation.

Initial interval	$F(x)$	$F^{\circ 2}(x)$	$F^{\circ 3}(x)$
$[1, 3]$	$(x + 5)/2$	$2x + 2$	$8x$
$[3, 4]$	$4x - 8$	$16x - 40$	$8x$
$[4, 8]$	$4x - 8$	$2x + 16$	$8x$

2.2 A small exact certificate of minimality

The orbit beginning at 1 is

$$1, 3, 4, 8, 24, 32, 64, 192, 256, \dots \quad (10)$$

In particular,

$$\det \begin{pmatrix} 1 & 3 & 4 \\ 3 & 4 & 8 \\ 4 & 8 & 24 \end{pmatrix} = -56 \neq 0. \quad (11)$$

Suppose $q_0x + q_1F(x) + q_2F^{\circ 2}(x) = 0$ on $\mathbb{R}$. Substitution of $x = 1, F(1), F^{\circ 2}(1)$ gives the matrix in (11) times $(q_0, q_1, q_2)^T$ equal to zero. Hence all three coefficients vanish. This rules out every nonzero relation of degree at most two. Relation (7) supplies a monic relation of degree three. Subtracting any other monic cubic relation would produce a relation of lower degree, so (8) is unique.

The roots of μ_F are

$$2, \quad -1 + i\sqrt{3}, \quad -1 - i\sqrt{3}. \quad (12)$$

They all have modulus 2. Deleting the nonreal pair leaves the polynomial $z - 2$, but $(z - 2)[F](1) = 1 \neq 0$. This is precisely the obstruction asked about in the selected problem. Degree three is also the smallest possible degree for a real polynomial to have both a real root and a nonreal conjugate pair.

2.3 The conjugacy behind the formula

The construction can also be understood without the itinerary table. On $[1, 8]$ define

$$h_0(u) = \begin{cases} 2u - 1, & 1 \leq u \leq 2, \\ u/2 + 2, & 2 \leq u \leq 4, \\ u, & 4 \leq u \leq 8. \end{cases} \quad (13)$$

Extend it by $H(8^k u) = 8^k h_0(u)$ for $u \in [1, 8]$, then extend oddly and set $H(0) = 0$. The map H is an increasing homeomorphism, and it commutes with $D_8(x) = 8x$. Direct calculation shows

$$F = H \circ D_2 \circ H^{-1}. \quad (14)$$

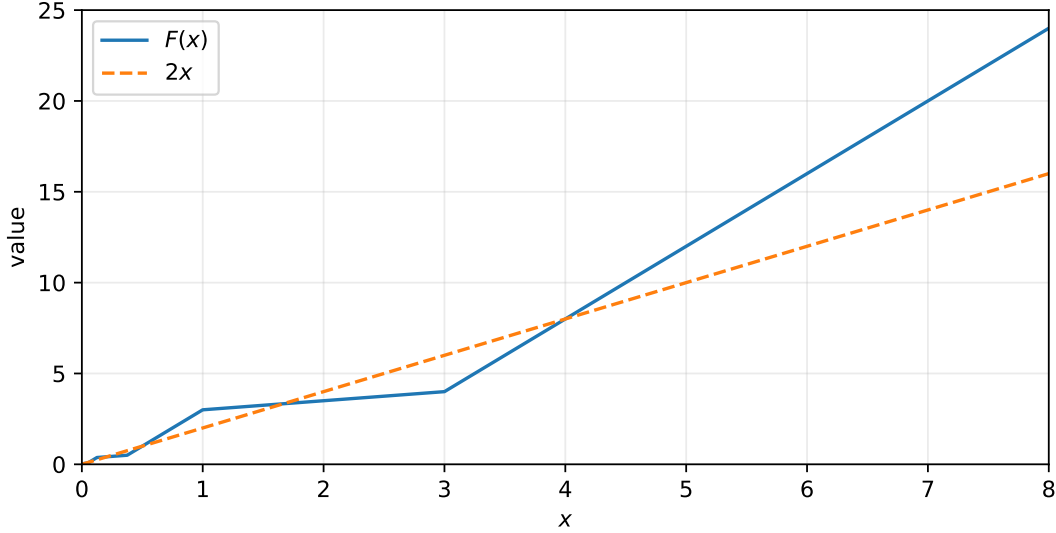


Figure 1: The exact cubic counterexample on one scale interval, compared with the homogeneous linear solution. There are infinitely many scaled copies of the two affine branches, accumulating at zero.

Consequently,

$$F^{\circ 3} = H \circ D_8 \circ H^{-1} = D_8. \quad (15)$$

The middle equality is useful only because H commutes with D_8 ; it need not commute with D_2 .

There is a general warning here. Arbitrary nonlinear conjugacy does *not* preserve an arbitrary linear combination of iterates. Although iterates conjugate one by one, a nonlinear conjugating map cannot be moved through a sum. In (14), commutation with the specific third iterate supplies the extra fact needed. In later constructions, we impose a linear difference equation on the conjugating coordinate itself instead.

2.4 A visible failure of differentiability

The scaled sequences $x_n = 8^{-n}$ and $y_n = 3 \cdot 8^{-n}$ both tend to zero, but

$$\frac{F(x_n)}{x_n} = 3, \quad \frac{F(y_n)}{y_n} = \frac{4}{3}. \quad (16)$$

Thus F is not differentiable at zero, despite being globally bi-Lipschitz. Section 9 explains why this is inevitable for a square-free circular spectrum of radius different from 1. The next example avoids this regularity issue altogether.

3 A real-analytic counterexample on the entire real line

Let

$$h(t) = t + \frac{1}{4} \sin \frac{\pi t}{2}, \quad f(x) = h(h^{-1}(x) + 1). \quad (17)$$

Here h^{-1} is the ordinary inverse function, not a reciprocal. Since

$$h'(t) = 1 + \frac{\pi}{8} \cos \frac{\pi t}{2} \quad \text{and} \quad 1 - \frac{\pi}{8} > 0, \quad (18)$$

h is strictly increasing. Also $|h(t) - t| \leq 1/4$, so it maps $\mathbb{R}$ onto $\mathbb{R}$. Its derivative never vanishes. The real-analytic inverse function theorem, applied at each real point, makes h^{-1} and f real analytic. No global complex-analytic inverse is asserted or required.

Theorem 3.1 (Analytic quartic counterexample). *The function in (17) is an increasing, globally bi-Lipschitz, real-analytic homeomorphism of $\mathbb{R}$. Its minimal iterative polynomial is*

$$\mu_f(z) = (z - 1)^2(z^2 + 1) = z^4 - 2z^3 + 2z^2 - 2z + 1. \quad (19)$$

It also satisfies $f^{\circ 4}(x) = x + 4$. The equation obtained by deleting the nonreal roots from (19) is not satisfied.

Proof. Conjugacy gives, for every integer n ,

$$f^{\circ n}(h(t)) = h(t + n) = t + n + \frac{1}{4} \sin \frac{\pi(t + n)}{2}. \quad (20)$$

The linear polynomial in n is killed by $(E - 1)^2$, where E shifts n to $n + 1$. The sine term is killed by $E^2 + 1$. The two operators commute, so their product kills the entire sequence. Therefore $\mu_f[f] = 0$. Periodicity of the sine also gives $h(t + 4) = h(t) + 4$, proving $f^{\circ 4}(x) = x + 4$.

The first orbit values at zero are

$$0, \quad \frac{5}{4}, \quad 2, \quad \frac{11}{4}, \quad 4, \quad \frac{21}{4}, \quad 6, \dots \quad (21)$$

Their 4×4 Hankel determinant is

$$\det \begin{pmatrix} 0 & 5/4 & 2 & 11/4 \\ 5/4 & 2 & 11/4 & 4 \\ 2 & 11/4 & 4 & 21/4 \\ 11/4 & 4 & 21/4 & 6 \end{pmatrix} = 1. \quad (22)$$

The same argument used for (11) rules out every relation of degree below four. This proves minimality.

After deleting i and $-i$, the remaining polynomial is $(z - 1)^2$. At $x = 0$ its residual is

$$f^{\circ 2}(0) - 2f(0) + 0 = 2 - 2 \cdot \frac{5}{4} = -\frac{1}{2}. \quad (23)$$

Thus the proposed deletion fails. Finally, $f'(h(t)) = h'(t + 1)/h'(t)$ is bounded above and bounded away from zero, which proves the bi-Lipschitz assertion. $\square$

All roots in (19) lie on the unit circle, and 1 is a real root of maximal modulus. This gives a second negative answer, now with real analyticity at every real point and with all polynomial coefficients nonzero. Notice that the map has no fixed point: $h(t + 1) > h(t)$. The distinction between a fixed point and a translation-like orbit will account for the different multiplicity rules at radius 1.

4 Orbit recurrences and minimal iterative polynomials

We now develop the tools for a general classification. The results in this section also explain why an exact orbit determinant is a valid minimality certificate.

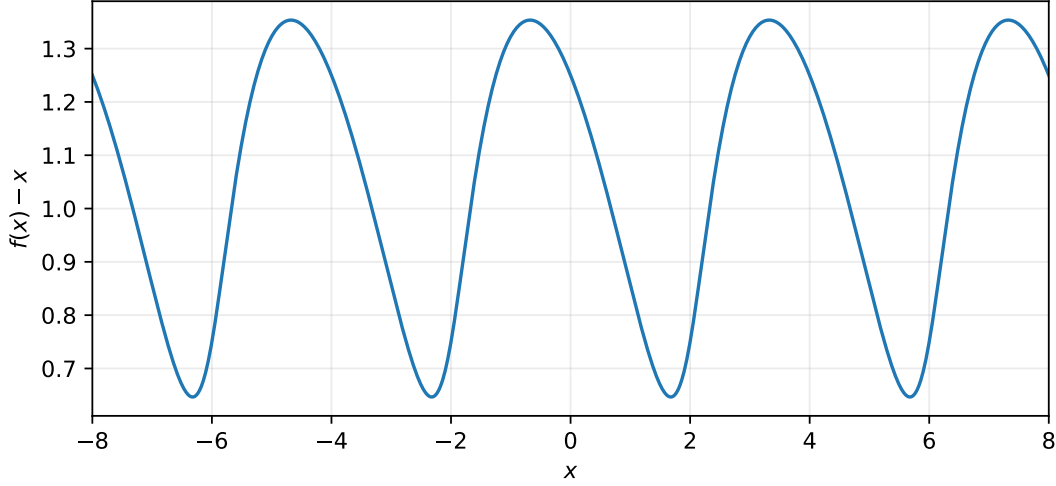


Figure 2: Numerical illustration of $f(x) - x$ for (17). The displacement is nonconstant and 4-periodic. The identities and the counterexample are established exactly in the text, not by this plot.

4.1 The annihilator ideal

Definition 4.1. For a function $f : \mathbb{R} \rightarrow \mathbb{R}$, let

$$\text{Ann}(f) = \{P \in \mathbb{R}[z] : P[f](x) = 0 \text{ for every } x \in \mathbb{R}\}. \quad (24)$$

If this set contains a nonzero polynomial, its monic polynomial of least degree is called the *minimal iterative polynomial* and is denoted by μ_f .

Lemma 4.2. *The set $\text{Ann}(f)$ is an ideal of $\mathbb{R}[z]$. When it is nonzero, $\text{Ann}(f) = (\mu_f)$.*

Proof. Closure under addition and scalar multiplication is immediate. For every polynomial P ,

$$(zP)[f](x) = P[f](f(x)). \quad (25)$$

Thus $P \in \text{Ann}(f)$ implies $zP \in \text{Ann}(f)$, and consequently every polynomial multiple of P belongs to the ideal. Divide any annihilator by the monic least-degree one. The remainder is another annihilator and has smaller degree, so it must be zero. $\square$

It is *right composition* in (25) that respects linear combinations. Applying f to a linear combination of its iterates would not give the same conclusion.

4.2 Why both forward and backward iterates exist

Lemma 4.3. *If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and $P[f] = 0$ for a polynomial with $P(0) \neq 0$, then f is a strictly monotone homeomorphism of $\mathbb{R}$.*

Proof. If $f(x) = f(y)$, then every positive iterate has the same value at x and y . Subtracting the two equations $P[f](x) = P[f](y) = 0$ gives $P(0)(x - y) = 0$, hence $x = y$. A continuous injective real function is strictly monotone.

If its image were a proper interval, at least one of its limits at $+\infty$ and $-\infty$ would be finite. Along the corresponding end, $f(x)$ would tend to a finite number, and continuity would make every positive iterate tend to a finite number as well. But the term $P(0)x$ in the equation would be unbounded. This contradiction proves surjectivity. A continuous strictly monotone bijection of $\mathbb{R}$ has a continuous inverse. $\square$

For such an f , define f^{on} for every $n \in \mathbb{Z}$ using inverse iterates when $n < 0$. For any starting point x , its orbit $u_x(n) = f^{on}(x)$ satisfies

$$\sum_{j=0}^d a_j u_x(n+j) = 0 \quad (n \in \mathbb{Z}). \quad (26)$$

Indeed, substitute $f^{on}(x)$ into $P[f] = 0$.

4.3 Exponential-polynomial solutions

Lemma 4.4 (Two-sided recurrence decomposition). *Let P be a polynomial with nonzero leading and constant coefficients. Write its distinct complex roots as λ , with multiplicities m_λ . Every complex sequence $u : \mathbb{Z} \rightarrow \mathbb{C}$ satisfying $P(E)u = 0$ has a unique representation*

$$u(n) = \sum_{\lambda} p_{\lambda}(n) \lambda^n, \quad \deg p_{\lambda} < m_{\lambda}. \quad (27)$$

For a real sequence, conjugate roots have conjugate coefficient polynomials. Its minimal recurrence polynomial is

$$\prod_{p_{\lambda} \neq 0} (z - \lambda)^{1 + \deg p_{\lambda}}. \quad (28)$$

Proof. For a polynomial p ,

$$(E - \mu)(p(n)\lambda^n) = \lambda^n(\lambda p(n+1) - \mu p(n)). \quad (29)$$

When $\mu = \lambda$, a nonconstant polynomial degree drops by one, and a constant is killed. When $\mu \neq \lambda$, the degree is preserved and the leading coefficient is multiplied by $\lambda - \mu$.

These observations prove that the proposed sequences satisfy the recurrence. They also prove their linear independence: in any putative linear relation, apply enough factors corresponding to all roots except one. The surviving root component is a nonzero polynomial times an exponential unless its coefficient polynomial was zero. There are $\deg P$ independent sequences. A two-sided recurrence with nonzero first and last coefficients is determined by $\deg P$ consecutive values, so these sequences form a basis. Formula (29) then proves the exact multiplicities in (28). $\square$

Lemma 4.5 (From individual orbits to the whole function). *The minimal iterative polynomial is the least common multiple of the minimal recurrence polynomials of all scalar orbits u_x . In particular, for each root, its multiplicity in μ_f is the maximum of the multiplicities with which it occurs in the scalar orbits.*

Proof. Each orbit is annihilated by μ_f , so its minimal recurrence polynomial divides μ_f . Conversely, any common multiple of all the orbit polynomials annihilates every orbit at $n = 0$, and hence annihilates f . Only finitely many roots and bounded multiplicities are possible, because they all occur in μ_f . Therefore their least common multiple and each stated maximum are attained using finitely many orbits. $\square$

This also gives a finite-dimensional interpretation. Right composition by f acts linearly on the span of $\text{id}, f, f^{\circ 2}, \dots$. The vector id is cyclic, and μ_f is the minimal polynomial on this cyclic space. The function f itself need not be linear as a map of real numbers.

5 The positivity lemma that controls a circular spectrum

A finite exponential polynomial on the unit circle is a sequence

$$v(n) = \sum_{\zeta \in S} p_{\zeta}(n) \zeta^n, \quad n \in \mathbb{Z}, \quad (30)$$

where S is a finite set of distinct complex numbers of modulus one. The sequence is real when the terms respect conjugation. The root $\zeta = 1$ contributes the nonoscillating polynomial part.

5.1 A recurrence fact for finite trigonometric sequences

Lemma 5.1. *For any finite set S on the unit circle, there are positive integers $N_k \rightarrow \infty$ such that $\zeta^{N_k} \rightarrow 1$ simultaneously for every $\zeta \in S$. Consequently, for a finite trigonometric sequence $T(n) = \sum_{\zeta \in S} c_{\zeta} \zeta^n$ and any fixed $\ell \in \mathbb{Z}$,*

$$T(N_k + \ell) \rightarrow T(\ell), \quad T(-N_k + \ell) \rightarrow T(\ell). \quad (31)$$

Proof. The vectors $(\zeta^n)_{\zeta \in S}$ lie in a compact finite product of circles. Choose a convergent subsequence with indices n_k whose successive gaps tend to infinity. The componentwise quotient of the vectors at indices n_{2k} and n_k tends to the identity vector. Set $N_k = n_{2k} - n_k$. The inverse quotients converge to the identity as well, giving both assertions in (31). $\square$

Lemma 5.2 (Nonnegative trigonometric sequences have positive mean). *Let $T(n) = \sum_{\zeta \in S} c_{\zeta} \zeta^n$ be real, nonnegative for every $n \in \mathbb{Z}$, and not identically zero. Then $1 \in S$ and $c_1 > 0$.*

Proof. The Cesàro mean of ζ^n over $n = 0, \dots, N-1$ tends to zero for $\zeta \neq 1$, by the finite geometric-sum formula. Thus the mean of T tends to c_1 , interpreted as zero if $1 \notin S$. Distinctness of the frequencies gives

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} |T(n)|^2 = \sum_{\zeta \in S} |c_{\zeta}|^2 > 0. \quad (32)$$

On the other hand, $0 \leq T(n) \leq M$ for some finite M , and $T(n)^2 \leq MT(n)$. Taking means proves $c_1 > 0$. $\square$

5.2 Parity and dominance of the nonoscillating term

Theorem 5.3 (Two-sided positivity). *Suppose the real sequence (30) is nonnegative for all $n \in \mathbb{Z}$ and is not identically zero. Let $d = \max\{\deg p_{\zeta} : p_{\zeta} \neq 0\}$. Then d is even. The polynomial p_1 is present, has degree exactly d , and has positive leading coefficient. The analogous statement for a nonpositive sequence holds after multiplication by -1 .*

Proof. Let c_{ζ} be the coefficient of n^d in p_{ζ} , with zero used when its degree is lower, and set

$$T(n) = \sum_{\zeta \in S} c_{\zeta} \zeta^n. \quad (33)$$

The leading trigonometric sequence is not identically zero, by the independence in Lemma 4.4. Uniformly as $|n| \rightarrow \infty$,

$$v(n) = n^d T(n) + O(|n|^{d-1}), \quad (34)$$

with the evident exact interpretation when $d = 0$.

At $+\infty$, nonnegativity gives $\liminf T(n) \geq 0$. Use the first sequence in (31) to conclude $T(\ell) \geq 0$ for every fixed integer ℓ . At $-\infty$, division by $|n|^d$ and the second return sequence give $(-1)^d T(\ell) \geq 0$ for every ℓ . If d were odd, T would vanish identically, a contradiction. Thus d is even. Lemma 5.2 now gives $c_1 > 0$. Hence p_1 has degree d and the asserted leading sign. $\square$

The use of both time directions is decisive. For a recurrence constrained only at nonnegative indices, an odd-degree polynomial may remain positive. A homeomorphism supplies all negative iterates, eliminating that possibility.

Corollary 5.4 (A vanishing unit-circle exponential polynomial). *If an exponential polynomial (30) tends to zero as $n \rightarrow +\infty$ or as $n \rightarrow -\infty$, it is identically zero.*

Proof. Choose its largest polynomial degree d and its leading trigonometric sequence T as in (34). The assumed limit implies $T(n) \rightarrow 0$ in the chosen direction. The corresponding return sequence in Lemma 5.1 gives $T(\ell) = 0$ for every ℓ , contrary to the choice of a nonzero leading degree. Thus a nonzero exponential polynomial cannot have the stated limit. $\square$

6 Classification of minimal polynomials on one circle

Call a polynomial *circular of radius r* if every one of its complex roots has modulus r . Multiplicities are counted in the usual algebraic sense. A missing root is assigned multiplicity zero.

Theorem 6.1 (Circular-spectrum classification). *Let $Q \in \mathbb{R}[z]$ be nonconstant and monic, and suppose all its roots lie on $|z| = r$ for some $r > 0$. Then Q is the minimal iterative polynomial of an increasing homeomorphism $f : \mathbb{R} \rightarrow \mathbb{R}$ if and only if the following conditions hold.*

- (i) *If $r \neq 1$, the positive real number r is a root of odd multiplicity m_r , and every other root has multiplicity at most m_r .*
- (ii) *If $r = 1$, either $Q(z) = z - 1$, or 1 is a root of even multiplicity $m_1 \geq 2$ and every other root has multiplicity at most $m_1 - 1$.*

In case (i), realizations can be chosen odd, globally bi-Lipschitz, and real analytic away from zero. In the nonidentity part of case (ii), realizations can be chosen globally bi-Lipschitz and real analytic at every real point; they have no fixed point.

This is a classification of the polynomials that can be *minimal*. It is not a parametrization of every function realizing any one polynomial. The minimality qualification is essential: for example, the identity satisfies $(z - 1)^7 [\text{id}] = 0$, but its minimal polynomial is only $z - 1$.

6.1 Necessity when $r \neq 1$

Let f be increasing and let $\mu_f = Q$. By Lemma 4.4, every orbit has the form

$$u_x(n) = r^n \sum_{|\zeta|=1} p_{x,\zeta}(n) \zeta^n. \quad (35)$$

If $r > 1$, this tends to zero as $n \rightarrow -\infty$; if $r < 1$, it tends to zero as $n \rightarrow +\infty$. Passing to this limit in $f(u_x(n)) = u_x(n+1)$ gives $f(0) = 0$. Since f and f^{-1} are increasing and fix zero, every nonzero orbit keeps the sign of its starting point for all $n \in \mathbb{Z}$.

For $x > 0$, divide (35) by $r^n > 0$ and apply Theorem 5.3. Its largest polynomial degree is even and is attained by the coefficient of $\zeta = 1$, corresponding to the characteristic root r . The same conclusion holds for $x < 0$ after a sign change. Thus, in every nonzero scalar orbit, the multiplicity of r is odd and is at least as large as every other multiplicity.

Now take the maxima over orbits using Lemma 4.5. The maximum of a nonempty finite set of odd multiplicities is odd. Every other global multiplicity is bounded by that maximum. This proves the necessity of condition (i).

6.2 Necessity on the unit circle

Suppose $r = 1$ and $f \neq \text{id}$. Choose a nonfixed starting point x . Because f is increasing, the difference sequence

$$v_x(n) = u_x(n+1) - u_x(n) \quad (36)$$

has a strict, constant sign for all $n \in \mathbb{Z}$.

Taking a first difference lowers the degree of the polynomial component at the root 1 by one. At every other root it preserves the degree, as (29) shows. Apply Theorem 5.3 to v_x , after a sign change if necessary. Its largest degree is even and is attained at 1. Therefore the polynomial component of u_x at 1 has odd degree $d \geq 1$, and every oscillatory component has degree at most $d - 1$. Equivalently, this orbit has an even multiplicity $d + 1$ at 1, strictly larger than all the other multiplicities.

The nonoscillating leading term dominates the orbit at both ends:

$$u_x(n) = an^d + O(|n|^{d-1}), \quad a \neq 0, \quad d \text{ odd}. \quad (37)$$

In particular, the two time tails go to opposite infinities. If f had a fixed point, an increasing orbit starting on either side of it could never cross it, contradicting (37). Hence a nonidentity f in this setting has no fixed point, and all its orbits are nonfixed. Lemma 4.5 now gives an even maximal multiplicity at 1 and the strict bound on every other multiplicity. If instead $f = \text{id}$, the minimal polynomial is $z - 1$. This proves condition (ii).

6.3 Encoding the remaining root directions

For the construction, write every root other than the distinguished positive root using an angle $0 < \theta_j \leq \pi$. An angle $0 < \theta_j < \pi$ denotes the conjugate pair $re^{i\theta_j}, re^{-i\theta_j}$, with common multiplicity m_j . The angle $\theta_j = \pi$ denotes the single real root $-r$, if present, with multiplicity m_j . Choose distinct angles and set

$$e_j = m_j - 1. \quad (38)$$

A term $r^{e_j} \cos(\theta_j t)$ then produces precisely the desired degree in its sampled orbit. At angle π , the two exponential expressions for the cosine coincide on integer samples and give the one real root, not two roots.

6.4 Construction for $r \neq 1$

Write $m_r = 2k + 1$ and set $a = \log r \neq 0$. Choose $C \geq 1$ so large that

$$\frac{k}{\sqrt{C}} \leq \frac{|a|}{2}, \quad (39)$$

and define

$$A(t) = (t^2 + C)^k, \quad W(t) = A(t) + \sum_j \varepsilon_j t^{e_j} \cos(\theta_j t), \quad h(t) = r^t W(t), \quad (40)$$

where every ε_j is positive and sufficiently small. The permitted multiplicities give $0 \leq e_j \leq 2k$. An explicit sufficient choice is any collection satisfying

$$\sum_j \varepsilon_j (|a| + e_j + \theta_j) \leq \frac{|a|}{4}. \quad (41)$$

When there are no other roots, the sum is empty.

Here are the estimates that justify the construction. For $0 \leq e \leq 2k$,

$$|t|^e \leq A(t), \quad e|t|^{e-1} \leq eA(t) \quad (e \geq 1), \quad \frac{|A'(t)|}{A(t)} \leq \frac{k}{\sqrt{C}}. \quad (42)$$

The first inequalities follow separately on $|t| \leq 1$ and $|t| \geq 1$. The last one follows by maximizing $2k|t|/(t^2 + C)$. Since (41) implies $\sum_j \varepsilon_j \leq 1/4$,

$$\frac{3}{4}A(t) \leq W(t) \leq \frac{5}{4}A(t). \quad (43)$$

Also

$$h'(t) = r^t (aW(t) + W'(t)). \quad (44)$$

The unperturbed expression $aA + A'$ has the sign of a and absolute value at least $|a|A/2$. Each perturbing expression

$$at^{e_j} \cos(\theta_j t) + e_j t^{e_j-1} \cos(\theta_j t) - \theta_j t^{e_j} \sin(\theta_j t) \quad (45)$$

has absolute value at most $(|a| + e_j + \theta_j)A(t)$, with the middle term omitted for $e_j = 0$. Thus

$$\operatorname{sgn} h'(t) = \operatorname{sgn} a, \quad \frac{|a|}{4} r^t A(t) \leq |h'(t)| \leq \frac{7|a|}{4} r^t A(t). \quad (46)$$

It follows that h is a real-analytic bijection from $\mathbb{R}$ onto $(0, \infty)$: it is increasing for $r > 1$ and decreasing for $r < 1$. Define

$$f(x) = h(h^{-1}(x) + 1) \quad (x > 0), \quad f(0) = 0, \quad f(-x) = -f(x). \quad (47)$$

Even when h is decreasing, the composition in (47) is increasing. Its limit at zero is zero, so the odd extension is an increasing homeomorphism of $\mathbb{R}$.

Every orbit on the positive half-line is $h(t + n)$. Expanding its cosine terms shows that it is annihilated by Q . Oddness handles negative orbits. More importantly, at the starting point $h(0)$ the sampled sequence is

$$h(n) = r^n \left((n^2 + C)^k + \sum_j \varepsilon_j n^{e_j} \cos(\theta_j n) \right). \quad (48)$$

The component at r has degree exactly $2k$, and every other component has exactly its assigned degree e_j , with nonzero leading coefficient. Lemma 4.4 proves that this single orbit has minimal polynomial Q . Hence $\mu_f = Q$.

Finally, (46) bounds $f'(h(t)) = h'(t+1)/h'(t)$ above and away from zero, because $A(t+1)/A(t)$ is positive, continuous, and tends to 1 at both ends. The resulting derivative bounds extend to secant bounds across zero. This proves the bi-Lipschitz and regularity assertions in case (i).

6.5 Construction on the unit circle

For the identity case choose $f = \text{id}$. Otherwise write $m_1 = 2k + 2$ and put

$$A(t) = \int_0^t (1 + s^2)^k ds = \sum_{\ell=0}^k \binom{k}{\ell} \frac{t^{2\ell+1}}{2\ell+1}. \quad (49)$$

Now A is an odd polynomial of degree $2k + 1$ and $A'(t) = (1 + t^2)^k > 0$. Since $e_j \leq 2k$, set

$$h(t) = A(t) + \sum_j \varepsilon_j t^{e_j} \cos(\theta_j t) \quad (50)$$

with all $\varepsilon_j > 0$ chosen so that

$$\sum_j \varepsilon_j (e_j + \theta_j) < \frac{1}{2}. \quad (51)$$

The same estimates as in (42), now relative to A' , imply

$$\frac{1}{2}(1 + t^2)^k < h'(t) < \frac{3}{2}(1 + t^2)^k. \quad (52)$$

The polynomial A has degree one more than every perturbing amplitude, so $h(t) \rightarrow \pm\infty$ as $t \rightarrow \pm\infty$. Therefore h is a real-analytic increasing diffeomorphism of $\mathbb{R}$.

Set $f = h \circ T \circ h^{-1}$, where $T(t) = t + 1$. All its orbits are $h(t + n)$ and are annihilated by Q . The orbit $h(n)$ contains a nonoscillating polynomial of degree $2k + 1$ and exactly the prescribed oscillating polynomial degrees e_j . Thus its minimal recurrence polynomial, and hence μ_f , is Q . The derivative ratio from (52) is bounded above and away from zero, proving global bi-Lipschitz continuity. The conjugacy also gives $f(x) > x$ everywhere and therefore no fixed point. This completes the sufficiency proof and the proof of Theorem 6.1.

7 Quantitative square-free realizations

The simplest part of the classification has a particularly transparent formula. It shows that survival of nonreal roots is not restricted to roots of unity or to an isolated cubic.

Fix $r > 0$, $r \neq 1$, and distinct angles $0 < \theta_j \leq \pi$. Choose nonzero real amplitudes ε_j and put

$$W(t) = 1 + \sum_{j=1}^q \varepsilon_j \cos(\theta_j t), \quad h(t) = r^t W(t). \quad (53)$$

Let $a = \log r$ and

$$B = \sum_{j=1}^q |\varepsilon_j| \sqrt{a^2 + \theta_j^2}. \quad (54)$$

Proposition 7.1. *If $B < |a|$, then the odd extension of $f = h \circ T \circ h^{-1}$ from $(0, \infty)$ to $\mathbb{R}$ is an increasing homeomorphism with minimal polynomial*

$$(z - r) \prod_{0 < \theta_j < \pi} (z^2 - 2r \cos \theta_j z + r^2) \prod_{\theta_j = \pi} (z + r). \quad (55)$$

It is real analytic away from zero, and its positive derivative on either open half-line satisfies

$$r \frac{|a| - B}{|a| + B} \leq f'(x) \leq r \frac{|a| + B}{|a| - B}. \quad (56)$$

These are also valid global lower and upper Lipschitz constants.

Proof. Since $B \geq |a| \sum_j |\varepsilon_j|$, the condition gives $W(t) > 0$. Moreover,

$$aW(t) + W'(t) = a + \sum_j \varepsilon_j (a \cos(\theta_j t) - \theta_j \sin(\theta_j t)) \quad (57)$$

differs from a by at most B . Thus h' has the sign of a , and its absolute value lies between $r^t(|a| - B)$ and $r^t(|a| + B)$. This proves that h is a suitable coordinate, and taking the ratio at $t + 1$ and t gives (56). Minimality follows from the nonzero frequency components in $h(n)$, exactly as before. $\square$

For example, taking one angle $0 < \theta < \pi$ yields a nonlinear solution whose minimal polynomial is

$$(z - r)(z^2 - 2r \cos \theta z + r^2). \quad (58)$$

The angle can be any such real number, not necessarily a rational multiple of π . In particular, $\theta = \pi/2$ and $r = 2$ give the cubic $z^3 - 2z^2 + 4z - 8$, all of whose coefficients are nonzero.

A small oscillatory component can coexist with strict monotonicity because the exponential envelope supplies a nonzero derivative margin. It is incorrect to infer that an orbit must lose every oscillatory mode merely because the unnormalized orbit is monotone. The cubic orbit (10) makes this visible:

$$\frac{F^{on}(1)}{2^n} = 1, \frac{3}{2}, 1, 1, \frac{3}{2}, 1, \dots \quad (59)$$

The normalized values oscillate, while the actual orbit is strictly increasing.

8 Sharp rigidity consequences for increasing solutions

The classification concerns minimal polynomials, but a given equation need not be minimal. Passing to divisors gives explicit rigidity tests.

Corollary 8.1 (Nonunit circle). *Let P be a nonconstant monic real polynomial whose roots lie on $|z| = r \neq 1$. If r is not a root, there is no increasing continuous solution of $P[f] = 0$ on $\mathbb{R}$. If r is a root, every increasing continuous solution is $f(x) = rx$ if and only if*

$$P(z) = (z - r)^m \quad \text{with } m \in \{1, 2\}. \quad (60)$$

Proof. Any such continuous solution is a homeomorphism by Lemma 4.3, and its minimal polynomial divides P . Theorem 6.1 requires the positive root r . Under (60), the only permitted odd multiplicity is one, so $\mu_f = z - r$.

Conversely, if P has any other root, take that real negative root or that nonreal conjugate pair together with a simple root at r . The square-free construction realizes the resulting divisor of P as a minimal polynomial of degree at least two; the realization is not homogeneous linear. If P has no other root but $m \geq 3$, the construction with $\mu_f = (z - r)^3$ supplies a nonlinear solution. Every solution in this nonunit setting fixes zero, so affine and homogeneous-linear rigidity coincide. $\square$

Corollary 8.2 (Unit circle). *Let P be nonconstant, monic, real, and supported on $|z| = 1$. Let M be the multiplicity of 1 in P .*

- (i) *If $M = 0$, there is no increasing continuous solution on $\mathbb{R}$.*
- (ii) *If $M \geq 1$, every increasing continuous solution is the identity if and only if $M = 1$.*
- (iii) *Among equations with an increasing solution, every increasing continuous solution is affine if and only if either $M = 1$, or $P(z) = (z - 1)^M$ with $M = 2$ or $M = 3$.*

Proof. If $M = 0$, the classification rules out every minimal divisor. If $M = 1$, the only possible minimal divisor is $z - 1$, regardless of the multiplicities of the other roots. When $M \geq 2$, every translation $x \mapsto x + c$ satisfies the equation, because its minimal polynomial for $c \neq 0$ is $(z - 1)^2$. This proves (i) and (ii).

If $P = (z - 1)^M$ with $M = 2$ or 3 , the only nonidentity minimal divisor allowed by the classification is $(z - 1)^2$. To see directly that its solutions are translations, write an orbit as $f^{\circ n}(x) = x + n(f(x) - x)$. For $x < y$, order preservation for every $n \in \mathbb{Z}$ gives

$$(y - x) + n((f(y) - y) - (f(x) - x)) > 0. \quad (61)$$

The coefficient of n must be zero, so $f(x) - x$ is constant.

If $M \geq 4$, a nonlinear realization of $(z - 1)^4$ is obtained from $h(t) = t + t^3/3$. If $M \geq 2$ and P has another root, adjoining that negative root or nonreal conjugate pair to $(z - 1)^2$ gives an admissible minimal divisor of degree greater than two. Its realization cannot be affine. Indeed, an affine increasing map with slope different from one has that positive slope as a characteristic root, whereas the only positive root here is one; an affine map of slope one is a translation and has minimal degree at most two. $\square$

These criteria make the radius-1 boundary especially clear. A simple positive root at 1 eliminates all other circular modes for increasing solutions. A double root at 1 supplies enough linear drift to allow nonconstant bounded oscillations. The analytic quartic example realizes exactly that first nonreal possibility.

9 What additional regularity does and does not restore

9.1 Differentiability is enough in the square-free nonunit case

Theorem 9.1. *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and continuous, and suppose its minimal iterative polynomial is square-free and supported on $|z| = r \neq 1$. If f is differentiable at zero, then $f(x) = rx$ for every x .*

Proof. We already know that $f(0) = 0$. Write $L = f'(0)$. Differentiating the iterative relation at zero, using the chain rule and the fixed point, gives $\mu_f(L) = 0$. Since an increasing differentiable map has $L \geq 0$, and zero is not a root, the only possible value is $L = r$.

For a fixed starting point, square-freeness gives $u(n) = r^n T(n)$, where T is a bounded finite trigonometric sequence. In the time direction in which $u(n) \rightarrow 0$, differentiability implies

$$u(n+1) - ru(n) = o(|u(n)|). \quad (62)$$

Divide by r^{n+1} and use boundedness of T to obtain $T(n+1) - T(n) \rightarrow 0$. This difference is a unit-circle exponential polynomial, so Corollary 5.4 makes it identically zero. In particular, $u(1) = ru(0)$. Since the starting point was arbitrary, $f = r \text{ id}$. $\square$

9.2 A positive-order Taylor remainder handles repeated roots

Theorem 9.2. *Let f be increasing and continuous, with a minimal iterative polynomial supported on $|z| = r \neq 1$, allowing repeated roots. Suppose f is differentiable at zero and, for some $\alpha > 0$,*

$$f(x) = f'(0)x + O(|x|^{1+\alpha}) \quad (x \rightarrow 0). \quad (63)$$

Then $f(x) = rx$ on $\mathbb{R}$. In particular, local $C^{1,\alpha}$ regularity at zero is sufficient.

Proof. As above, $f(0) = 0$ and $f'(0) = r$. Each orbit has the form $u(n) = r^n v(n)$, where v is a unit-circle exponential polynomial and $|v(n)| = O((1+|n|)^d)$ for some d . Apply (63) in the time direction in which $r^n \rightarrow 0$. It gives

$$v(n+1) - v(n) = \frac{u(n+1) - ru(n)}{r^{n+1}} \quad (64)$$

$$= O\left(r^{\alpha n} (1+|n|)^{d(1+\alpha)}\right) \rightarrow 0. \quad (65)$$

The last limit holds both for $r < 1, n \rightarrow +\infty$ and for $r > 1, n \rightarrow -\infty$. By Corollary 5.4, the difference vanishes identically. Thus $u(1) = ru(0)$ for every starting point. $\square$

9.3 C^1 alone is not sufficient with repeated roots

Consider the particularly simple coordinate

$$h(t) = e^t(t^2 + 4), \quad h'(t) = e^t((t+1)^2 + 3) > 0. \quad (66)$$

Use $f = h \circ T \circ h^{-1}$ on $(0, \infty)$ and extend it oddly through zero. Its minimal iterative polynomial is $(z - e)^3$. For $x = h(t)$,

$$\frac{f(x)}{x} = e \frac{(t+1)^2 + 4}{t^2 + 4} \rightarrow e \quad (t \rightarrow -\infty). \quad (67)$$

Also $f'(h(t)) = h'(t+1)/h'(t) \rightarrow e$ as $t \rightarrow -\infty$. Thus the extension is C^1 at zero and consequently C^1 on $\mathbb{R}$. It is nonlinear, because its minimal polynomial has degree three.

For every $\alpha > 0$, however,

$$\frac{|f(h(t)) - eh(t)|}{h(t)^{1+\alpha}} = \frac{e|2t+1|e^{-\alpha t}}{(t^2+4)^{1+\alpha}} \rightarrow \infty \quad (t \rightarrow -\infty). \quad (68)$$

So the positive-order remainder hypothesis of Theorem 9.2 genuinely strengthens bare differentiability. The analytic quartic of Section 3, on the other hand, shows that even real analyticity cannot restore a universal elimination rule at radius one, where the map need not have any fixed point.

10 Exact verification and reproducible artifacts

10.1 What the exact program checks

The script `code/verify.py` requires only Python 3.9 or later and its standard library. All algebraic checks use `Fractions.Fraction`; there is no floating-point arithmetic in these certificates. It implements the cubic function in two independent ways: the direct two-branch scaled formula and the polygonal conjugacy (14).

The recorded run tested 4,026 distinct rational starting points, including positive and negative points, multiple scales, and exact breakpoints. At every point it checked the cubic identity, oddness, scaling, agreement of the two implementations, and inverse identities. It also checked 4,025 adjacent exact secant slopes against the bounds $1/2$ and 4 . These finite tests supplement the all-real branch proof; they are not its logical replacement.

The program separately verifies recurrences on the integer range $-80 \leq n \leq 80$ and computes nonzero exact Hankel determinants for four sample orbits. The symbolic exponential-polynomial arguments explain why the recurrences hold for every integer and why the determinants certify the stated minimal orders.

Table 2: Exact sample certificates. The companion JSON file records the full matrices and polynomial coefficients.

Minimal polynomial	Degree	Hankel determinant
$z^3 - 8$	3	-56
$(z - 1)^2(z^2 + 1)$	4	1
$(z - 2)^3$	3	-512
$(z - 2)^5(z + 2)^4(z^2 + 4)^3$	15	nonzero exact integer

For the degree-15 example the sampled coordinate is

$$h(n) = 2^n \left((n^2 + 64)^2 + \frac{n^2}{1024} \cos \frac{\pi n}{2} + \frac{n^3}{1024} \cos \pi n \right). \quad (69)$$

Every sampled trigonometric value is an integer, so the whole sequence is rational. The real-variable version satisfies the derivative bounds in Section 6.4. Indeed, $k = 2$, $C = 64$, $1/2 < \log 2 < 1$, and $\pi < 4$ give

$$\frac{k}{\sqrt{C}} = \frac{1}{4} < \frac{\log 2}{2}, \quad \frac{2 \log 2 + 5 + 3\pi/2}{1024} < \frac{13}{1024} < \frac{\log 2}{4}. \quad (70)$$

Thus this is not merely a recurrence sequence detached from a monotone function; the construction realizes it as an actual orbit.

10.2 Numerical illustration, clearly separated

The optional script `code/analytic_demo.py` evaluates (17) by bisection. Since $|h(t) - t| \leq 1/4$, the inverse of x is bracketed by $x - 1/4$ and $x + 1/4$ before iteration begins. On 2,001 points in $[-10, 10]$, the recorded double-precision run gave a maximum absolute quartic residual of approximately $7.11 \cdot 10^{-15}$ and a maximum absolute residual in $f^{\circ 4}(x) = x + 4$ of approximately $1.24 \cdot 10^{-14}$. These numbers are implementation checks, not error bounds valid for all inputs and not part of the proof.

The plotting option generates the two vector figures used in this paper. The exact script does not require a plotting library. The article's counterexample remains independently checkable using the three-row itinerary table and the small integer or rational determinants.

11 Conclusions and the remaining boundary

The selected root-elimination question has a definite negative answer under its stated general hypotheses. The self-similar cubic has a real root and a nonreal pair at the same maximal modulus, yet every one of those roots is required by a minimal relation. The analytic quartic shows that this conclusion is not an artifact of a nonsmooth example.

The circular-spectrum classification identifies the mechanism more precisely. Away from radius one, positivity of the normalized two-sided orbit requires an even maximal polynomial degree, producing an odd multiplicity at the positive root. On the unit circle, it is the orbit *increment* that must have constant sign; taking the increment lowers the degree at one, producing an even multiplicity there and a strict multiplicity margin over every oscillating mode. Small trigonometric perturbations realize all and only the permitted patterns.

Several boundaries remain outside this report. We have not classified minimal polynomials with roots on several distinct circles. We have not classified decreasing homeomorphisms, for which consecutive iterates reverse order. Nor have we handled noninjective equations with zero constant coefficient or arbitrary proper interval domains. These are limitations of the proved statements, not qualifications needed by the two counterexamples on $\mathbb{R}$.

Finally, mathematical verification and historical priority are different questions. The paper gives complete arguments and exact finite certificates; it does not supply an independent referee report or an exhaustive priority search. The broader linearity problem is not declared solved. What is established here is the failure of the proposed general nonreal-root deletion and an exact, constructive classification in the increasing single-circle setting.

A Source and status audit

The literature search was performed on 20 September 2026. The primary sources below were used to identify the question and delimit the claims, not to replace the proofs in this article.

The target. Draga–Morawiec [1], Problem 6.1, is present in the retrieved HTML and in the rendered page 13 of arXiv:1503.00570v2. The preprint has 14 pages. Its published bibliographic record is *Aequationes Mathematicae* 90 (2016), 935–950, with DOI 10.1007/s00010-016-0420-4. The source is about preserving the solution equation after reducing the characteristic polynomial, rather than merely preserving the existence of at least one solution.

Subsequent context. Draga’s note [2] gives another order-reduction theorem and mentions the unrestricted nonreal-root issue in its final remarks. The later paper *Means of iterates* [4] treats a particular family of iterative mean equations and uses characteristic-root analysis. A 2021 paper [5] studies a class of fourth-order equations under restrictions on the characteristic roots. These sources show continuing work on the subject; none of the retrieved material established priority for the counterexample developed here.

Initial lead. Petrov’s question [3] was the initial candidate. Its displayed answer count was zero in the retrieved page. An unanswered question page is not, by itself, proof that no published solution exists elsewhere.

Search limitation. The search included the authors’ names, polynomial-like iterative equations, nonreal-root elimination, equal-modulus roots, and the numbered problem. Some returned results were irrelevant and were not used. The search was not a complete citation-database survey. Thus the report does not attach a first-discovery claim to the elementary construction or to the classification theorem. The complete source identifiers and this status distinction are also recorded in `notes/literature_audit.md`.

B Reproduction, selection, and exclusion record

From the archive’s top-level directory, the following commands reproduce the exact certificates and, with the optional plotting dependency, the numerical illustrations:

```
python code/verify.py
python code/analytic_demo.py --plots
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The supplied figure files allow the PDF to be compiled without running the plotting script. No bibliography processor is required.

The exact verifier normalizes positive rational numbers by multiplication or division by eight, not by taking an approximate logarithm. This avoids wrong branch choices at powers of eight. Its independent conjugacy implementation also uses rational operations. The recurrence polynomials in the JSON certificate are stored with coefficients in *ascending* order, so, for example, $z^3 - 8$ is represented by `[-8, 0, 0, 1]`.

The random area draw was:

Number of listed areas	80
Generator call	<code>secrets.randbelow(80)</code>
Returned zero-based index	17
Selected one-based index	18
Selected area	Functional equations
Redraws	0

An entropy-backed draw is recorded rather than seed-replayed; running a new draw is not expected to reproduce the same index. The full 80-element list is in `data/selection.json`.

The excluded input was the 1,095-line file `manifest(1).tex`, with SHA-256 fingerprint

`729fb6e3aef3f30c7c5284c4296681fba72b6d6af68211f178ad0bc3c5d15b41.`

The file was reviewed as an exclusion list; this report does not certify the correctness or priority of its individual mathematical claims. A suggested new manifest entry is supplied separately so that future selections can exclude the present target as well.

References

- [1] S. Draga and J. Morawiec, *Reducing the polynomial-like iterative equations order and a generalized Zoltán Boros' problem*, *Aequationes Mathematicae* **90** (2016), 935–950. doi:10.1007/s00010-016-0420-4. Preprint: arXiv:1503.00570v2, 1 December 2015. Problem 6.1 is on preprint page 13.
- [2] S. Draga, *A note on the order of polynomial-like iterative equations*, *Commentationes Mathematicae* **56** (2016), 243–249. Preprint title: *A note on the polynomial-like iterative equations order*, arXiv:1604.01287v2, 14 November 2016.
- [3] F. Petrov, *Continuous linear recurrent relations*, MathOverflow question 232493, 1 March 2016. <https://mathoverflow.net/questions/232493/continuous-linear-recurrent-relations>. Retrieved 20 September 2026.
- [4] S. Draga and J. Morawiec, *Means of iterates*, arXiv:1801.05006v2, 19 February 2018.
- [5] S. Thangroongvongthana, V. Laohakosol, and S. Mavecha, *A class of continuous solutions of a fourth order polynomial-like iterative equation*, *Current Applied Science and Technology* **21**(3) (2021), 495–523. <https://li01.tci-thaijo.org/index.php/cast/article/view/248735>.